NADIM MAHMUD

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PROFESSIONAL SUMMARY

I am a Software Engineer focused on building scalable, event-driven systems and integrating AI microservices into production applications. I specialize in the Python/Django ecosystem, with deep experience in asynchronous architectures using Celery, Redis, and WebSockets. Beyond standard REST APIs, I enjoy solving complex performance bottlenecks, optimizing SQL/NoSQL databases (PostgreSQL, MongoDB), and automating cloud deployments. Coming from a background of solving 600+ competitive programming problems, I bring a strong analytical mindset and a genuine passion for writing clean, testable code in collaborative engineering teams.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript
- **Frameworks:** Django, DRF, React, Redux
- **Frontend & Styling:** HTML, CSS, Tailwind, Bootstrap
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite
- **Brokers & Cache:** Redis, RabbitMQ, Celery
- **System Architecture & Design:** Microservices, Event-Driven Architecture, Message Brokers, Pub/Sub, CQRS, API Gateway, System Optimization
- **Backend Concepts:** RESTful API, WebSocket, Server-Sent Events (SSE), JWT & RBAC, API Caching, Throttling, Asynchronous Programming, AI Microservices
- **Tools & Testing:** Git, Docker, VPS, AWS, CI/CD, Postman, Pytest

PROFESSIONAL EXPERIENCE

Software Engineer | *Confidential Tech Startup – Dhaka, Bangladesh* *Oct 2025 – Present*

- Designed, developed, and maintained scalable RESTful backend services using Python and Django REST Framework for AI-powered business applications.
- Integrated third-party APIs and internal AI microservices; implemented JWT-based authentication and role-based access control for secure, production-ready systems.
- Optimized system performance, scalability, and reliability through Redis-based API caching, reducing average API response latency and improving overall throughput.
- Built reliable, non-blocking backend workflows using asynchronous programming and Celery/Redis background task processing, integrated with machine learning pipelines.
- Achieved 85% code coverage by writing comprehensive unit and integration tests using Pytest, adhering to Test-Driven Development (TDD) principles to ensure the stability of critical backend processes.
- Set up automated CI/CD pipelines using GitHub Actions and collaborated with an 11-person cross-functional engineering team for seamless Docker and AWS-based cloud deployments, ensuring 99.9% application uptime.

Team Leader | *Vcube Soft and Tech Limited – Dhaka, Bangladesh* *Jun 2019 – Oct 2024*

- Promoted from Operator to Project Quality Checker, Mentor, and Team Leader through consistent performance.
- Managed team coordination and quality assurance for 2D/3D building floor plan projects.
- Mentored and led a cross-functional team of 6 members, maintaining productive communication across remote and in-office environments and significantly reducing onboarding time for new hires.

PROJECTS

AI-Powered Insurance App

- Built a scalable insurance management platform featuring a freemium model supporting 10,000+ potential users: free-tier users interact with a real-time AI chatbot via WebSocket for insurance guidance, while premium subscribers access full case management with document submission and AI-driven, document-aware guidance.
- Managed WebSocket session flow and routed structured conversation data to an internal AI microservice for context-aware chat responses; handled connection management and session-based conversation state.
- Implemented manual email-based registration, Firebase-powered social authentication (Google, Apple), in-app purchase subscription system, role-based access control, JWT-based authentication with API throttling, cursor-based pagination for chat history, and Celery/Redis background tasks for automated weekly insurance claim reminders via Firebase Push Notifications.

Eat At Home

- Meal tracking and nutrition analytics platform; integrated Plaid API to fetch real bank transaction history for accurate restaurant spending tracking and automated food expense categorization.
- Called AI microservice to analyze food photos and return pricing estimates; implemented Redis caching for AI responses, decreasing latency by over 50%, and integrated results into weekly meal planning logic and Celery-based automated scheduling.
- Implemented manual email-based registration alongside Firebase-powered social authentication (Google, Facebook, Apple); built scalable backend using Django ORM, class-based views, serializers, and routers, using PostgreSQL for analytics data storage. Integrated WebSockets for live updates, Redis for background tasks, and Firebase for real-time push notifications.

Animal Service

- Full-featured pet marketplace built to handle 5,000+ daily active users; implemented Redis caching for marketplace queries, improving page load times by 40% for listings and vet appointments.
- Implemented proximity-based transport matching via Google Maps API — users search within a defined radius, nearby drivers receive real-time push notifications (integrated via Firebase and OneSignal), and a single driver accepts the request.
- Integrated Telr payments, digital wallet for seller earnings/withdrawals, and built real-time 1-on-1 chat using WebSocket; managed asynchronous background processes using Redis. Admin panel included with fraud prevention, verification workflows, and optimized pagination for large datasets.
- Implemented multi-method registration (email and phone), role-based access control for buyers, sellers, admins, veterinary doctors, and transport providers; optimized PostgreSQL schema across marketplace, bookings, medical records, and transactional workflows.

EDUCATION

Bangladesh Sweden Polytechnic Institute

Diploma in Computer Technology | Rangamati, Bangladesh *Jan 2014 – Jun 2018*

KEY ACHIEVEMENTS

- Solved 600+ programming problems across Codeforces, LeetCode, HackerRank, CodeChef, and InterviewBit.
- Secured 38th place in the weekly CodeChef contest by solving all problems in June 2025.
- Achieved 11th position in the Inter Polytechnic Programming Contest at IUBAT in 2016.